In [ ]:
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 EXPERIMENT 6: DATA VISUALIZATION WITH SEABORN

PIN:241UDS0097
 Aim :-
To visualize and analyze a dataset using Seaborn plots including scatter plots, KDE, histograms, count plots, bar plots, boxplots, strip plots, joint plots, and heatmaps.
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In [ ]:
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## Step 1: Import the required libraries and dataset
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In [1]:
```
# Import the libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [ ]:
```
# Load the dataset (keep the CSV in the same folder as this notebook)
electricity_data_241UDS0097 =
pd.read_csv("Experiment_6_241UDS0097_Electricity_Tariff.csv")
```

In [ ]:
```
# Info of the dataset
electricity_data_241UDS0097.info()
```

In [ ]:
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# Display the first five rows
electricity_data_241UDS0097.head()
```

In [ ]:
```
# State-wise data points in the dataframe
electricity_data_241UDS0097.value_counts("state")
```

In [ ]:
```
# Convert date column to datetime format
electricity_data_241UDS0097["date"] =
pd.to_datetime(electricity_data_241UDS0097["date"])
# Extract year for the heatmap analysis
electricity_data_241UDS0097["year"] =
electricity_data_241UDS0097["date"].dt.year
```

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## Step 2: Visualize data using various plots
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In [ ]:
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### Scatter Plot – Electricity Tariff Across States Over Time
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In [ ]:
```
plt.figure(figsize=(15, 6))
sns.scatterplot(x="date", y="rate", hue="state",
data=electricity_data_241UDS0097, alpha=0.8)
plt.title("Electricity Tariff Comparison Across States")
plt.xlabel("Date")
plt.ylabel("Tariff (practice units)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

In [ ]:
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### Histograms and KDE — Distribution of Tariff Values
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In [ ]:
```
plt.figure(figsize=(10, 4))
sns.kdeplot(data=electricity_data_241UDS0097, x="rate", color="teal")
plt.title("Electricity Tariff Distribution with KDE")
plt.xlabel("Tariff (practice units)")
plt.show()
```

In [ ]:
```
plt.figure(figsize=(10, 4))
sns.histplot(data=electricity_data_241UDS0097, x="rate", bins=30, kde=True,
color="purple")
plt.title("Electricity Tariff Distribution with Histogram + KDE")
plt.xlabel("Tariff (practice units)")
plt.show()
```

In [ ]:
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### Count Plot — Number of Data Points for Each State
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In [ ]:
```
plt.figure(figsize=(7, 4))
sns.countplot(x="state", data=electricity_data_241UDS0097, hue="state",
palette="viridis", legend=False)
plt.title("Number of Data Points per State")
plt.xlabel("State")
plt.ylabel("Count")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```

In [ ]:
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### Barplot — Average Tariff per State
```

In [ ]:
```
plt.figure(figsize=(7, 4))
sns.barplot(x="state", y="rate", data=electricity_data_241UDS0097,
hue="state", estimator=np.mean, palette="Set1", errorbar=None, legend=False)
plt.title("Average Electricity Tariff per State")
plt.xlabel("State")
plt.ylabel("Average tariff (practice units)")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```

In [ ]:
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### Boxplot — Distribution and Potential Outliers
```

In [ ]:
```
plt.figure(figsize=(7, 4), dpi=100)
sns.boxplot(x="rate", y="state", data=electricity_data_241UDS0097,
hue="state", palette="Set2", legend=False)
plt.title("Boxplot of Electricity Tariff by State")
plt.xlabel("Tariff (practice units)")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```

In [ ]:
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### Strip Plot — Tariff Distribution by State
```

In [ ]:
```
plt.figure(figsize=(10, 4), dpi=120)
sns.stripplot(x="state", y="rate", data=electricity_data_241UDS0097,
hue="state", palette="Set1", alpha=0.5, legend=False)
# Alternative shown in the source experiment: sns.swarmplot(...)
plt.title("Stripplot of Electricity Tariff by State")
plt.xlabel("State")
plt.ylabel("Tariff (practice units)")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()
```

In [ ]:
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### Joint Plot — Distribution of Tariff vs Date
```

In [ ]:
```
sns.jointplot(x="date", y="rate", data=electricity_data_241UDS0097,
height=7)
plt.suptitle("Jointplot: Tariff vs Date", y=1.02)
plt.show()
```

In [ ]:
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### Heatmap — Average Tariff per Year per State
```

In [ ]:
```
# Average tariff per year per state
yearly_avg_241UDS0097 = electricity_data_241UDS0097.groupby(["year",
"state"])["rate"].mean()
yearly_avg_241UDS0097
```

In [ ]:
```
# Reshape so each row is a year and each column is a state
yearly_avg_241UDS0097 = electricity_data_241UDS0097.groupby(["year",
"state"])["rate"].mean().unstack()
yearly_avg_241UDS0097.head()
```

In [ ]:
```
plt.figure(figsize=(8, 7), dpi=100)
sns.heatmap(yearly_avg_241UDS0097, cmap="YlGnBu", fmt=".2f", annot=True)
plt.title("Yearly Average Electricity Tariff by State")
plt.xlabel("State")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```